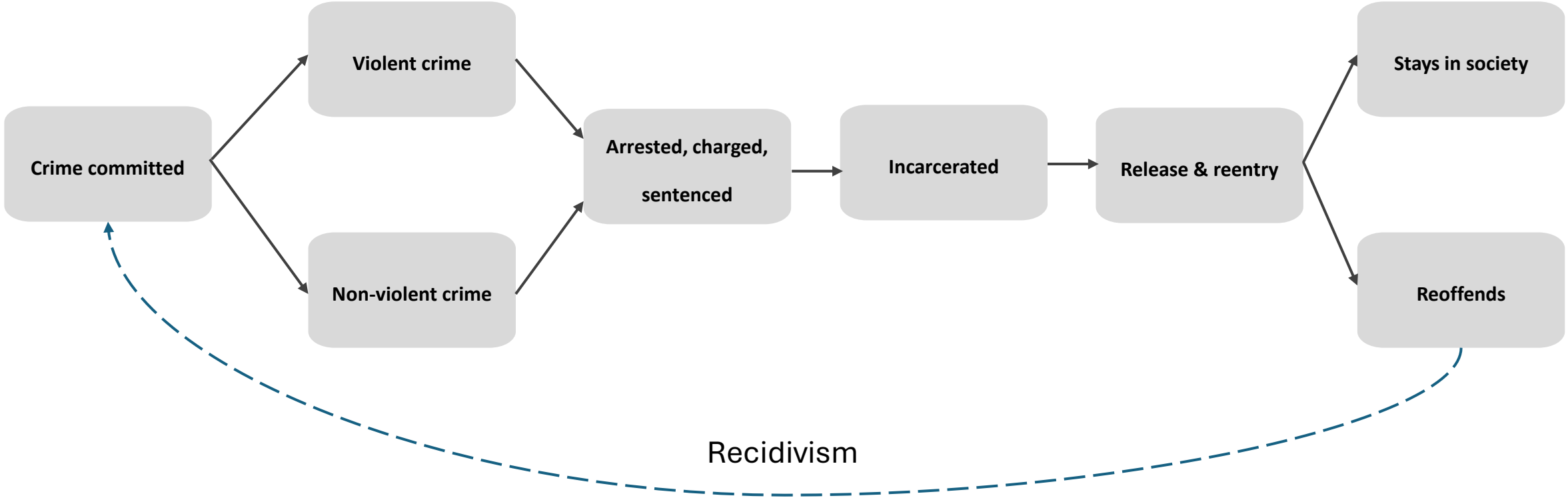
# Recidivism Prediction in the Criminal Justice System

Michelle Zheng

MSDS696

April 2026

# Recidivism & Reentry Cycle




- Can criminal history records be used to construct predictive models that accurately and equitably identify individuals at elevated risk of recidivism following release from incarceration? Can the fairness score surpass the existing algorithm COMPAS?

# Data Set

- From ProPublica – public data in Broward County, Florida from 2013 - 2014
- >18,000 records of criminal defendants
  - Demographics
  - Criminal history
  - COMPAS scoring outputs
- Target Variable: is_recid (binary, re-arrested < 2 years)
  - 56% re-arrested
  - 46% stayed in society
- Evaluate fairness within race

| Race | Count | Distribution |
|---|---|---|
| African-American | 1,941 | 53% |
| Caucasian | 1,199 | 33% |
| Hispanic | 322 | 9% |
| Other | 175 | 5% |
| Asian | 16 | <1% |
| Native American | 11 | <1% |

# Data Cleaning & EDA

- Numeric missing values imputed with median
- Categorical missing values filled as 'Missing' category
- COMPAS score column dropped as part of comparison baseline
- Name & date of birth columns dropped for privacy

| Type | Variable Examples |
|---|---|
| Numeric | Age |
| Numeric | Juvenile History Charge Count |
| Numeric | Start/End |
| Numeric | ... (10 total) |
| Categorical | Age Category |
| Categorical | Sex |
| Categorical | Charged Description |
| Categorical | ... (5 total) |

# Feature Engineering

| Feature Engineering Category | Data Table Columns | New Feature Name & Description | Reasoning |
|---|---|---|---|
| Age | age, age_cat | age_bucket (0-21, 21-25, 25-30, 30-40, 40+) | Research describes that criminal activity is more likely to occur in young adults & lowers with age. |
| | | is_young_adult (<25) | |
| Juvenile History | juv_fel_count, juv_misd_count | total_juv_charges | Summing gives one indicator of total juvenile charges & indicates if they do have a juvenile history |
| | | has_juv_history (binary) | |
| Prior Offenses | priors_count | priors_log1p (log transform) | During EDA, most defendants show average of 3 priors, while small fraction of data set has more than 3. Transforming compresses the scale of prior distribution |
| | | is_repeat_offender (≥3) | |
| | | lifetime_charges | |
| Interaction | priors_count × is_young_adult | priors_x_young (interaction term) | Defendants in young adult age group with prior counts are at the most risk of re-offending. |

# Models from Standard Scalar for Prediction

| Model | Parameter Description | ROC-AUC | Acc | F1 | Prec | Recall | FPR | FNR | DPD |
|---|---|---|---|---|---|---|---|---|---|
| **Extra Trees** | 200 trees | **0.8901** | 0.8054 | 0.8051 | 0.7987 | 0.7963 | 0.1861 | 0.2037 | 0.2979 |
| **LightGBM** | 200 estimators | **0.8850** | 0.8103 | 0.8101 | 0.7982 | 0.8106 | 0.1900 | 0.1894 | 0.2708 |
| **XGBoost** | 200 estimators | **0.8838** | 0.8086 | 0.8083 | 0.8021 | 0.7993 | 0.1828 | 0.2007 | 0.3214 |
| Random Forest | 200 trees | 0.8724 | 0.7840 | 0.7838 | 0.7707 | 0.7844 | 0.2164 | 0.2156 | 0.4643 |
| Gradient Boosting | 200 estimators | 0.8517 | 0.7740 | 0.7740 | 0.7479 | 0.7999 | 0.2500 | 0.2001 | 0.3392 |
| MLP Neural Network | Layers: 120→64, max_iter=500 | 0.8184 | 0.7557 | 0.7552 | 0.7502 | 0.7381 | 0.2280 | 0.2619 | 0.2589 |
| Decision Tree | max_depth=8 | 0.8160 | 0.7423 | 0.7423 | 0.7139 | 0.7749 | 0.2880 | 0.2251 | 0.2869 |
| Logistic Regression | max_iter=1000 | 0.8155 | 0.7469 | 0.7467 | 0.7301 | 0.7518 | 0.2577 | 0.2482 | 0.3008 |
| K-Nearest Neighbors | k=7 | 0.8113 | 0.7349 | 0.7346 | 0.7218 | 0.7304 | 0.2610 | 0.2696 | 0.5268 |
| Naive Bayes | Gaussian Naive Bayes | 0.7738 | 0.6057 | 0.5368 | 0.8262 | 0.2286 | 0.0446 | 0.7714 | 0.2232 |
| COMPAS for Comparison | — | 0.70 | 0.65 | — | — | — | — | — | ~0.20 |

**Fairness measure:**

- DPD: Demographic Parity Difference, equal Positive Prediction Rate across groups

# Outcome & Future Work

- All models substantially outperform COMPAS on predictive accuracy. The top 3 models surpass COMPAS by approximately 19–20 percentage points.
  1. Extra Trees (0.890)
  2. LightGBM (0.885)
  3. XGBoost (0.884)

- Higher accuracy does not guarantee lower racial bias, there is accuracy-fairness trade-off. The top 2 performing models exhibit higher DPD values than several lower-accuracy alternatives.
- Top 2 vs Last 2:
  - Extra Trees (best AUC) has DPD=0.298
  - LightGBM has DPD=0.271
  - MLP Neural Network  DPD=0.259
  - Naive Bayes achieves DPD=0.223 (approaching COMPAS's benchmark) but at the cost of a 77% false negative rate that renders it operationally useless.
- Optimizing purely for predictive accuracy tends to amplify existing social disparities encoded in the training data

- Future Work: Apply SHAP to identify which features drive individual predictions and if race variables (charge type distributions that correlate with race due to policing patterns) are driving predictions.

# References

Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016). Machine Bias: There's Software Used Across the Country to Predict Future Criminals. And It's Biased Against Blacks. ProPublica. https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing
Barocas, S., Hardt, M., & Narayanan, A. (2019). Fairness and Machine Learning: Limitations and Opportunities. fairmlbook.org.
Chouldechova, A. (2017). Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments. Big Data, 5(2), 153–163.
Dressel, J., & Farid, H. (2018). The Accuracy, Fairness, and Limits of Predicting Recidivism. Science Advances, 4(1).
Northpointe Inc. (2012). Practitioner's Guide to COMPAS Core. Traverse City, MI: Northpointe Institute for Public Management.